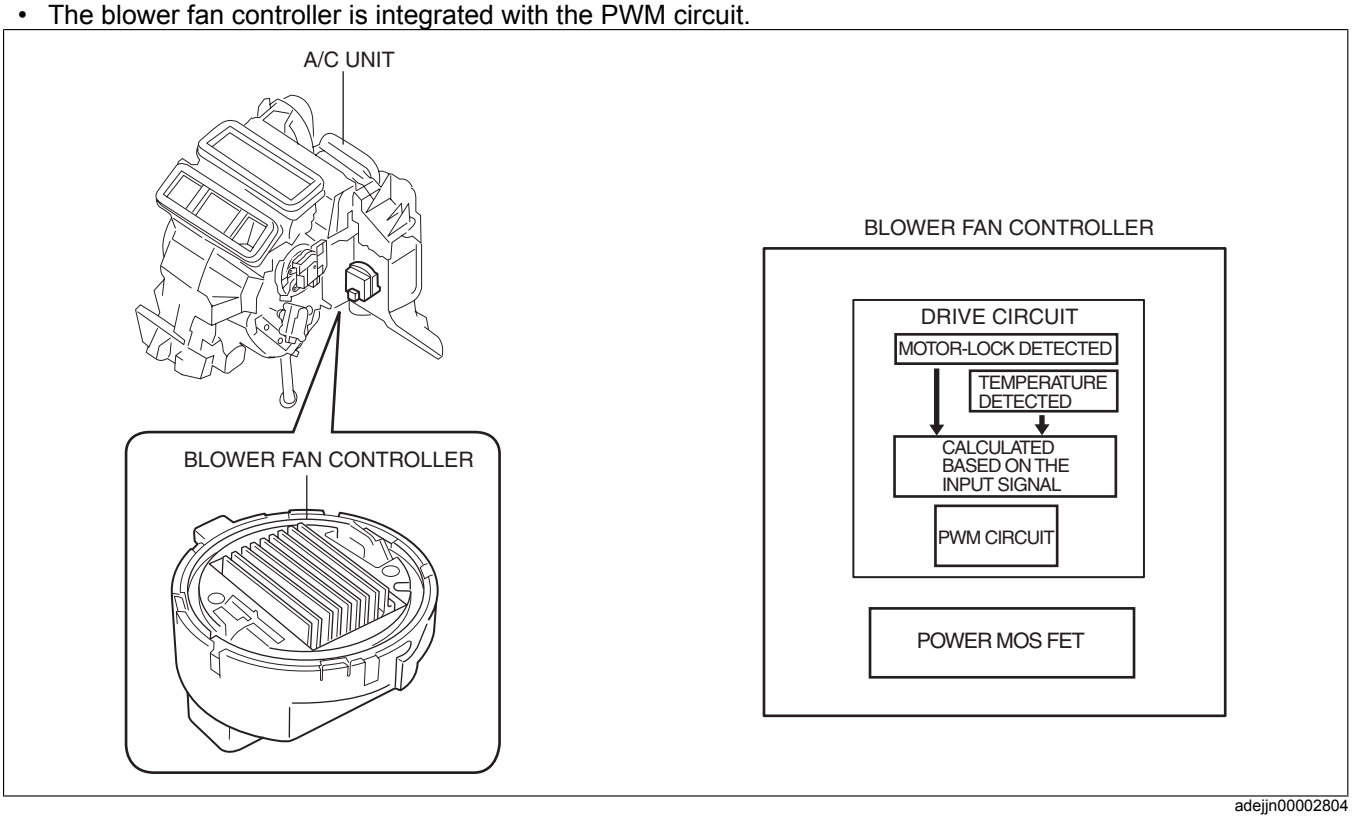
BLOWER FAN CONTROLLER

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Purpose
- The blower fan controller controls the blower motor rotation speed.

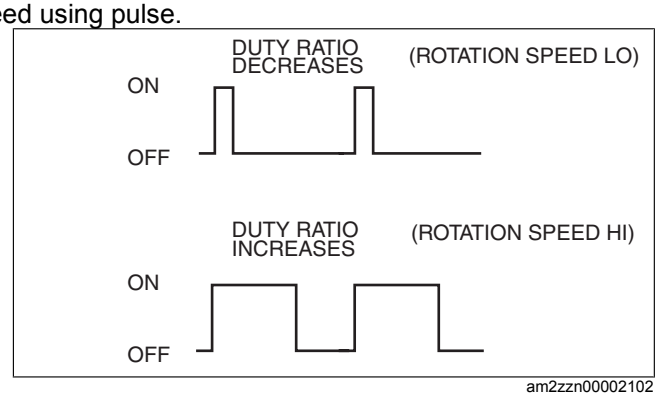
Function
- The blower fan controller controls the built-in power MOS FET using pulse signals and changes the blower motor operational voltage.
- By placing the fan in the upper part of the blower fan controller, temperature increase in the internal circuit is suppressed.
- The blower fan controller is equipped with a fail-safe function for the purpose of protecting the internal circuits.

Construction
- The blower fan controller is positioned in the A/C unit.
- The blower fan controller is integrated with the PWM circuit.



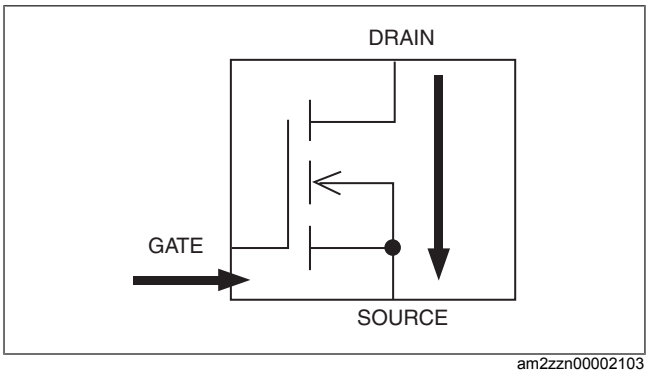
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PWM (pulse width modulation) circuit
- The PWM circuit controls the blower motor rotation speed using pulse.
- The PWM circuit determines the duty ratio (pulse ON time) and turns the power MOS FET on/off at high speed.



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- When voltage of the threshold or more is applied to the gate, a large amount of electricity flows between the drain and source and the power MOS FET turns on.
- The rotation of the blower motor is fast when the power MOS FET ON period is long (duty ratio is high), and it is slow when the period is short (duty ratio is low).
- The PWM circuit performs duty control using the power MOS FET ON or OFF operation to reduce electric power loss and suppress heat generation in the circuit.



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Operation
- For the blower fan controller operation, refer to the BLOWER SYSTEM. (See BLOWER SYSTEM.)

Fail-safe
- The blower fan controller stops output for a certain period of time when it detects the following conditions:
  - — Motor lock
    - The blower motor locks and over-current flows.
  - — Abnormal temperature
    - The power MOS FET increases temperature abnormally due to half-lock (over-current of a level such that motor-lock is not detected) of the blower motor.